

Three claim levels

Running mathematics stays in prose: the scan word e_n visits the 2^n leaves in tree order, and its cost telescopes because $\sum_{i < 2^n} v_2(i) < 2^n$. No box — the reader is not being asked to audit anything yet.

LEAN $\vdash$ Adic.Dyadic.streaming

Streaming is amortized $O(1)$

A full left-to-right scan of the 2^n leaves costs at most $4 \cdot 2^n$: the Euler word succeeds from the root, returns to the root, and visits exactly the leaves, in order, once each.

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theorem streaming (n : Nat) :
  cost (euler n) ≤ 4 * 2 ^ n ∧
  run (euler n) (rootHead n) = some (rootHead n) ∧
  leafVisits (euler n) (rootHead n) = some (leftToRightLeaves n) ∧
  (leftToRightLeaves n).length = 2 ^ n
axioms: propext, Quot.sound  hash: demo-hand-pasted
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DESK-PROVED — not mechanized

Mergesort's honest cost

On D , mergesort over N elements of w bits costs $\Theta(Nw \log N)$ bit-operations: each of the $\log N$ passes is a constant number of interleaved streams. Desk-argued; mechanization queued behind the ladder.

UNFORMALIZED

No cost-bounded cone

Any family of maps presenting one object as the limit of the projection chain $\{\mathbb{Z}/2^n\}$ has unbounded cost, while the colimit is presented at cost zero per element. (Letter 2's thesis sentence, awaiting a cost-enriched setting even to state.)